

AWS GENERATIVE AI PROFESSIONAL

2. Data, Embeddings, and Retrieval-Augmented Generation

Detailed tutorial with metaphor, worked architecture case, design-first contract, failure laboratory, and certification practice

BUSINESS INVARIANT

Every repair recommendation must cite an approved source applicable to the exact asset and effective date.

Certification alignment: Domain 2 - Data for Generative AI

Tutorial promise: You will move from concept to an architecture decision, then prove the decision through contracts, tests, operational evidence, and scenario questions.

1. Learning Map and Mental Model

RAG improves an answer only when the retrieval layer returns relevant, current, permitted evidence. Chunking, metadata, embedding selection, indexing, query transformation, reranking, and citation validation form a data product, not a one-time upload task. Architects must preserve lineage from source document to chunk, embedding, retrieved passage, generated claim, and user response. Evaluation therefore separates retrieval quality from generation quality.

PLAIN ENGLISH

RAG is an open-book examination. A clever student still performs badly if the librarian hands over the wrong chapter, an expired manual, or somebody else's confidential notes.

MEMORY HOOK

Adding more documents without metadata is not knowledge management; it is giving the librarian a leaf blower.

Learning outcomes

- Explain the topic to a nontechnical stakeholder without losing the control boundary.
- Identify the accountable actor, protected business truth, and prohibited AI action.
- Select AWS capabilities from requirements instead of starting with a service list.
- Design the normal workflow and the safe-failure workflow.
- Specify measurable evidence that can accept or reject the solution.

Core vocabulary

Term	Meaning in this tutorial
Business outcome	The measurable value the organization expects.
Invariant	Every repair recommendation must cite an approved source applicable to the exact asset and effective date.
Probabilistic component	A model output that may vary even when the request appears similar.
Deterministic boundary	Code, policy, schema, or approval that produces a repeatable enforceable decision.
Evidence contract	Recall@k, precision@k, citation precision, freshness SLA, source lineage, and retrieval latency.

2. Step-by-Step Architecture Tutorial

Begin with intent, then introduce technology only when a requirement needs a capability.

1	Frame the business problem A manufacturer assists technicians with repair guidance. Approved manuals and bulletins enter S3 through a versioned ingestion workflow. Metadata filters constrain plant, product, asset class, region, and validity date before Bedrock generates cited steps. Conflicting documents trigger engineering review.
2	Freeze the invariant Every repair recommendation must cite an approved source applicable to the exact asset and effective date.
3	Define authority The accountable human or deterministic service owns the final business state. The AI assists only within this rule: Treat document governance, metadata quality, and retrieval evaluation as release gates.
4	Design the interface contract Specify identity, required fields, schemas, idempotency, accepted and rejected responses, status transitions, and audit identifiers before implementation.
5	Separate fast intake from variable work Choose synchronous processing only for bounded validation and durable acceptance. Use asynchronous coordination when latency, bursts, or retries vary. Candidate services: Amazon S3, AWS Glue, Amazon OpenSearch Serverless, Bedrock Knowledge Bases, AWS Step Functions, CloudWatch
6	Add deterministic gates Validate authorization, schema, policy, calculations, citations, safety, and allowed state transitions outside the model.
7	Define safe failure Quarantine failed ingestion and return a controlled no-answer when evidence conflicts.
8	Accept with evidence Recall@k, precision@k, citation precision, freshness SLA, source lineage, and retrieval latency.

3. Worked Case Study

CASE CONTEXT

A manufacturer assists technicians with repair guidance. Approved manuals and bulletins enter S3 through a versioned ingestion workflow. Metadata filters constrain plant, product, asset class, region, and validity date before Bedrock generates cited steps. Conflicting documents trigger engineering review.

Architecture reasoning

Question	Professional decision
What is business truth?	Every repair recommendation must cite an approved source applicable to the exact asset and effective date.
Where does AI add value?	Treat document governance, metadata quality, and retrieval evaluation as release gates.
What must AI never own?	Authorization, identity, official calculations, irreversible state changes, and final release authority unless the contract explicitly assigns and proves them.
How does the system fail?	Quarantine failed ingestion and return a controlled no-answer when evidence conflicts.
What proves readiness?	Recall@k, precision@k, citation precision, freshness SLA, source lineage, and retrieval latency.

End-to-end walkthrough

- The actor authenticates and submits a schema-valid request with a correlation identifier and idempotency key.
- The API enforces authorization and business preconditions before recording durable acceptance.
- The workflow obtains only authorized, versioned context and calls the selected model within latency, token, and safety budgets.
- Deterministic controls verify structure, grounding, calculations, permissions, and allowed state transition.
- The system records PASS, REVIEW, RETRY, or REJECT with evidence and exposes the status to the actor.
- An accountable owner reviews exceptions; monitoring detects threshold breaches and can stop or roll back the release.

4. Design-First Contract and Test Plan

Plain-English contract

WHEN	An authenticated, authorized actor submits a request that satisfies the published input contract.
THE SYSTEM PROMISES	Treat document governance, metadata quality, and retrieval evaluation as release gates.
THE SYSTEM REFUSES	Every repair recommendation must cite an approved source applicable to the exact asset and effective date.
IF SOMETHING FAILS	Quarantine failed ingestion and return a controlled no-answer when evidence conflicts.

Acceptance tests

Test	Injection or stimulus	Pass condition
Contract validation	Remove a required field or change its type.	Rejected before model invocation or side effect; reason is traceable.
Authorization	Use another role or tenant identity.	Access denied; no protected context is retrieved or disclosed.
Idempotency	Replay the same request or event.	One logical business result; replay returns the known state.
AI quality	Return schema-invalid, unsupported, or unsafe output.	Deterministic gate blocks release and routes RETRY, REVIEW, or REJECT.
Dependency failure	Throttle or time out an AWS or model dependency.	Quarantine failed ingestion and return a controlled no-answer when evidence conflicts.
Evidence	Recall@k, precision@k, citation precision, freshness SLA, source lineage, and retrieval latency.	Every threshold is observable, attributable, and linked to a release.

5. Operations, Evidence, and Well-Architected Review

Pillar	Questions to ask
Operational excellence	Can operators trace a request, reproduce the decision, run the failure playbook, and roll back?
Security	Is identity enforced before retrieval and tool use? Are secrets, data, and logs protected?
Reliability	Are retries bounded and idempotent? Does every accepted item reach a terminal state?
Performance efficiency	Which latency component dominates? Is the selected model and retrieval pattern appropriate?
Cost optimization	Is cost measured per accepted outcome? Are tokens, caching, batching, and capacity governed?
Sustainability	Is the smallest capable model used? Does the design scale to zero or process efficiently when idle?

Evidence dashboard

Layer	Measure	Why it matters
Business	Outcome KPI and protected invariant	Proves the system creates value without violating business truth.
Application	Acceptance latency, error rate, terminal-state completion	Separates user experience from downstream AI processing.
AI quality	Grounding, factuality, safety, variance	Measures probabilistic quality directly.
Security	Denied access, leakage tests, suspicious prompts	Shows the authority boundary remains intact.
Operations	Queue age, retries, DLQ, alarm response, rollback time	Shows safe recovery rather than silent failure.
Economics	Cost per accepted outcome and model utilization	Prevents cheap tokens from masking expensive failed work.

6. Certification-Style Knowledge Check

Choose the best architectural response. Answers are on the next page.

1. Which design choice BEST protects the business requirement in this tutorial?

- A. Allow the model to make the final decision when it sounds confident.
- B. Treat document governance, metadata quality, and retrieval evaluation as release gates.
- C. Optimize speed first and add authority controls after launch.
- D. Share cached answers across users without authorization.

2. The AI dependency times out after the business request is accepted. What should the system do?

- A. Delete the request and ask the user to retry from memory.
- B. Mark the workflow successful without processing.
- C. Repeat every side effect without an idempotency key.
- D. Quarantine failed ingestion and return a controlled no-answer when evidence conflicts.

3. Which evidence package MOST strongly demonstrates production readiness?

- A. One successful demonstration.
- B. Recall@k, precision@k, citation precision, freshness SLA, source lineage, and retrieval latency.
- C. A fluent response screenshot.
- D. A claim that the largest model should be reliable.

4. A generated response contains an unsupported but plausible claim. What is the BEST action?

- A. Publish it because users probably will not notice.
- B. Raise the temperature and regenerate creatively.
- C. Block the business side effect and route the result through the defined review or failure path.
- D. Remove the citation requirement.

5. What is the correct interpretation of these services: Amazon S3, AWS Glue, Amazon OpenSearch Serverless, Bedrock Knowledge Bases, AWS Step Functions, CloudWatch?

- A. Managed services remove the need for business ownership.
- B. The model should dynamically grant its own IAM permissions.
- C. They implement capabilities, while contracts, deterministic controls, tests, and accountable owners establish trust.
- D. Their use guarantees zero hallucinations.

7. Answers, Explanations, and Teaching Prompts

1. Correct answer: B

The architecture should treat document governance, metadata quality, and retrieval evaluation as release gates. This directly protects the invariant: Every repair recommendation must cite an approved source applicable to the exact asset and effective date.

Key knowledge: Business authority and invariant

Teaching prompt: Ask which business fact, authority, or evidence would be lost if the selected control were removed.

2. Correct answer: D

The documented safe-failure response is: Quarantine failed ingestion and return a controlled no-answer when evidence conflicts. Accepted state must not disappear because a probabilistic dependency is unavailable.

Key knowledge: Safe failure and durable state

Teaching prompt: Ask which business fact, authority, or evidence would be lost if the selected control were removed.

3. Correct answer: B

The required evidence is measurable and repeatable: Recall@k, precision@k, citation precision, freshness SLA, source lineage, and retrieval latency.

Key knowledge: Measurable evidence

Teaching prompt: Ask which business fact, authority, or evidence would be lost if the selected control were removed.

4. Correct answer: C

Unsupported output is a controlled failure. Grounding and deterministic validation must prevent it from becoming business truth.

Key knowledge: Grounding and deterministic validation

Teaching prompt: Ask which business fact, authority, or evidence would be lost if the selected control were removed.

5. Correct answer: C

AWS services provide capabilities, scaling, and operational mechanisms. The business contract still defines authority, safe failure, and measurable acceptance.

Key knowledge: Managed services within a governed architecture

Teaching prompt: Ask which business fact, authority, or evidence would be lost if the selected control were removed.

Final recap

- Business invariant: Every repair recommendation must cite an approved source applicable to the exact asset and effective date.
- Architecture decision: Treat document governance, metadata quality, and retrieval evaluation as release gates.
- Safe failure: Quarantine failed ingestion and return a controlled no-answer when evidence conflicts.
- Acceptance evidence: Recall@k, precision@k, citation precision, freshness SLA, source lineage, and retrieval latency.

EXAM TIP

Prefer the option that satisfies the explicit requirement, preserves authority, fails safely, minimizes unnecessary operational burden, and can be proven with measurable evidence.

Original teaching material. Verify current AWS service documentation and the current certification exam guide before production or examination decisions.